Position, distance, displacement

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish a location, a total path length, and a change in position.

SUCCESS CRITERIA - I can...

1. Label the origin, start, turn and finish.
2. Calculate path distance and signed displacement separately.
3. Explain why position and displacement can differ.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “position” and “distance” to compare your model or result. Sentence starters are optional.

position: location measured from a chosen origin

distance: total length of the path traveled

displacement: change from initial position to final position, with direction

01 Bring back an earlier idea

A cart is at 3, 8, 13 m at 0, 1, 2 s.
Predict its position at 7 s.
State an assumption and a measurement that could challenge it.

Sentence starter: I predict ... . My model supports this because ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $\Delta x = x_f - x_i$; distance = sum of path lengths. Choose an origin and positive direction. i = initial; f = final.

Worked reasoning: A walker starts at 2 m, reaches 6 m, then ends at 4 m. Distance=4+2=6 m. Displacement=4-2=2 m right.

Your turn: Draw the two route arrows and a separate start-to-finish arrow.
Label the length of each.

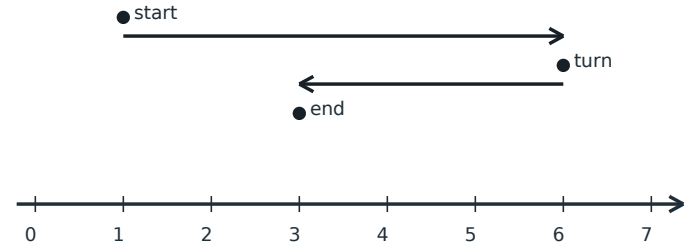
Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Read the route

The route starts at 1 m, goes to 6 m, and ends at 3 m.
Label start, turn, and end on the picture.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Position (m); right is positive

Marzano check - circle: ① ② ③

04 Count the whole path

For the route in 3, find each leg and the total distance.
First leg: ____ m. Return leg: ____ m. Total: ____ m.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Connect start to finish

For the same route, draw one arrow from start to end. **Give** the displacement and final position. Are they the same number?

Support: I end ____ m to the ____ of where I started.

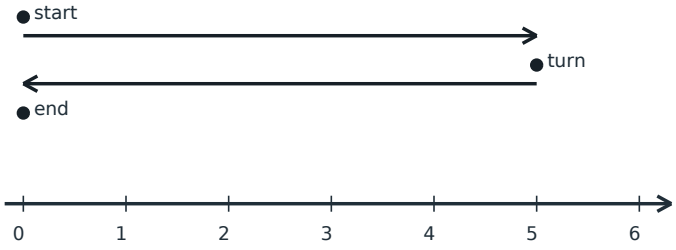
Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: 1 2 3

06 A complete trip back

Start at 0 m. Walk 5 m east, then 5 m west.
Find distance and displacement.
Explain why one is zero.

Sentence starter: My claim is … . The diagram/data show …, so … .



Position (m); right is positive

Marzano check - circle: 1 2 3

07 Fix the wording

A student says, “My displacement was 8 m because I walked 8 m altogether.” Must that be true?
Use a short route or sketch to explain.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

08 Choose a method

A cart is observed at 3, 8, 13 m, one second apart.
Record two patterns and point to values that support each.

Sentence starter: I notice … . This supports … because … .

Time (s)	Position (m)
0	3
1	8
2	13

Marzano check - circle: 1 2 3

09 Exit check

Start at 2 m, move to 7 m, and finish at 5 m.
Find total distance, displacement with direction, and final position.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

10 STRETCH: Apply it another way

Design two routes that start at 1 m and end at 4 m: one of distance 3 m and one of distance 7 m.
Draw and check both.

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: 1 2 3